Creating ASP.NET Core Web API Project in Visual Studio

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Creating an ASP.NET Core Web API Project in Visual Studio

In this article, I will discuss **how to create, build, run, and test the ASP.NET Core Web API Project in Visual Studio**. Please read our previous article discussing How to Create, Build, Run, and Test the ASP.NET Core Web API project using .NET Core in Linux Ubuntu Server.

Visual Studio is a cross-platform, lightweight, but very simple lightweight, fast, and cross-platform services that contains rapidly, productive, and powerful tools and features. We can easily create **Using Visual Studio .NET 2017**, Visual Studio 2019 and Visual Studio 2022 Preview, and Visual Studio Code. Visual Studio Code is a lightweight, fast, and cross-platform code editor. We can easily create a new Web API project with ease using Visual Studio. Visual Studio Code is a lightweight, fast, and cross-platform code editor. We can easily create a new Web API project with ease using Visual Studio.

Opening Visual Studio and Creating a New Project

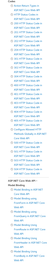
When you open Visual Studio, then the first screen you will see is **"Create a new project"**. This is the starting point for creating any kind of .NET project. To begin building a new Web API, click on the **"Create a new project"** option as shown in the image below.

Key Points:

- This is the entry point for starting any new solution in Visual Studio.
- Clicking this option lets Visual Studio find you quick set up a fresh project workspace.
- It's the first step in the process of building a new project template depending on whether you're building a console app, desktop app, web service, or library.
- Behind the scenes, Visual Studio prepares to generate the required folder structure, project files, and configuration settings.

Choosing the ASP.NET Core Web API Template

After selecting Create a new project, the **Create a new Project window** appears, as shown in the image below, listing various project templates. Since we are building a Web API, we need to select the appropriate template. Visual Studio offers several Web API templates, including those for ASP.NET Core, ASP.NET Framework, and Azure Functions. Choose the template that aligns with your development goals. For this tutorial, we'll use the ASP.NET Core Web API template, which is designed for building modern, cross-platform Web APIs using ASP.NET Core. This template provides a lightweight, fast, and cross-platform foundation for building Web APIs that can be deployed on various platforms, including Windows, Linux, and macOS.



Key Points:

- The template sets up all the necessary dependencies for building RESTful APIs.
- It saves you time by automatically including routing, error, program configuration, and controller examples.

Configuring Your New Project

Once you select the appropriate template, Visual Studio prompts you to configure your new project in the **Configure your new project** screen, where you specify:

- **Project Name**: Provide a name for the Web API project.
- **Location**: Choose where you want the project to be saved on your system.
- **Solution Name**: You can keep the same as the project or use a broader solution name if you plan to add multiple projects.

Naming your project properly helps in solution organization and ease, especially in enterprise applications. Once filled, click the **Next** button, as shown in the image below.

Key Points:

- Clear naming and organization ensure easy navigation and maintenance.
- Consider how this project will integrate with other components in your ecosystem.

Additional Information Window

After you click the Next button, the Additional Information window will open. Here you configure:

- **Target Framework**: Select .NET 8.0 (LTS), an extensive long-term support and stability, .NET 9.0 (Preview) leading-edge features, or Microsoft Azure Functions v4 for cloud-ready deployments.
- **Authentication Type**: For now, select **None**, as we are not implementing login or user management. If you choose an option, Visual Studio will automatically add the necessary NuGet packages and configuration templates.
- **Other Options**:
  - **Configure for HTTPS**: Enables secure communication.
  - **Enable Docker**: Adds a Dockerfile for cross-platform containerization.
  - **Enable OpenAPI support**: Generates API documentation.
  - **Use controllers**: Uses traditional controllers instead of minimal APIs.
  - **Do Not Use Top-Level Statements**: Keeps the traditional Program.cs structure.

Click on the **Create** button. Visual Studio will create the ASP.NET Core Web API project with the following file and directory structure. It can easily navigate, edit, and manage these files and folders in detail.

Click on the Icon **Show all files** in Solution Explorer, as shown in the image below.

Understanding the Options

- **Configure for HTTPS**: This option configures your ASP.NET Core application to use HTTPS (Hypertext Transfer Protocol Secure) for secure communication. It ensures that all data exchanged between the client and server is encrypted, safeguarding the application's data and HTTPS endpoints.
- **Enable Docker**: Adds a Dockerfile to your project, facilitating containerization. Docker simplifies deployment across various environments, ensuring consistency and ease of deployment.
- **Enable OpenAPI Support**: Adds Swagger UI for easy API testing and documentation. Swagger automatically generates interactive documentation, allowing developers to test API endpoints directly from the browser.
- **Do Not Use Top-Level Statements**: A new feature in ASP.NET Core 6 and later, this option allows you to write your application's startup code in a more concise, streamlined format. If you prefer the traditional Program.cs structure with explicit namespaces and classes, uncheck this option. This option is useful for those who prefer a more structured approach.

Project File & Folder Structure

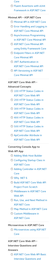
Once you click on the **Create** button, Visual Studio will create the ASP.NET Core Web API project with the following file and directory structure. It can easily navigate, edit, and manage these files and folders in detail.

Project Run & Initial Output

Now, click on the **Run** button, Visual Studio will create the ASP.NET Core Web API project with the following file and directory structure. It can easily navigate, edit, and manage these files and folders in detail.

Project Build

Now, click on **Build** → **Build Solution** as shown in the image below.



The will compile all projects in the solution and show the build status in the Output window.

Project Run Options

Option 1: Using Visual Studio Menu

Find the Visual Studio Toolbar, click **Build** → **Build Solution** as shown in the image below.

Option 2: Right-click and Build

This method allows you to quickly build a specific project within a solution containing multiple projects. If you right-click on the solution and choose **Build Solution**, all projects within it will be compiled.

Option 3: Use Keyboard Shortcut

- Press **Ctrl + Shift + B**

This is the fastest way to trigger a build for the entire solution.

Note: The Build Solution command compiles everything. However, if you only want to build a single project, use the right-click method instead.

Project Run Options for ASP.NET Core Web API Application in Visual Studio:

Step 1: Select Launch Profile

Visual Studio provides multiple ways to run your API:

- **IIS Express**
- **WebAPIDemo(Kestrel)**
- **Docker**
- **WSL (Windows Subsystem for Linux)**

Select **Kestrel** when the project is ready to run, as shown in the image below.

Debug
Any Debug Target
Any Target
Any Target
Any